# DENVER RTD EAGLE P3
# COMMUNICATION SYSTEM

# FIELD MODIFICATION INSTRUCTIONS
# FMI #13: SOFTWARE UPDATE

| | |
|---|---|
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# Record of Revisions

| Rev. | Date | Description | Entered By |
|---|---|---|---|
| 1.0 | 9/20/2016 | Initial release | D. S. Jang |
| | | | |
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# 1. INTRODUCTION

This FMI provides instructions to install the Communication System update to the vehicles.

# 2. SUMMARY

This FMI applies to following software configuration item:

- CRA-PAC SW change.

# 3. INSTRUCTIONS

## 3.1 LOCATION OF EQUIPMENT

Following figures show the location of equipment to be updated.

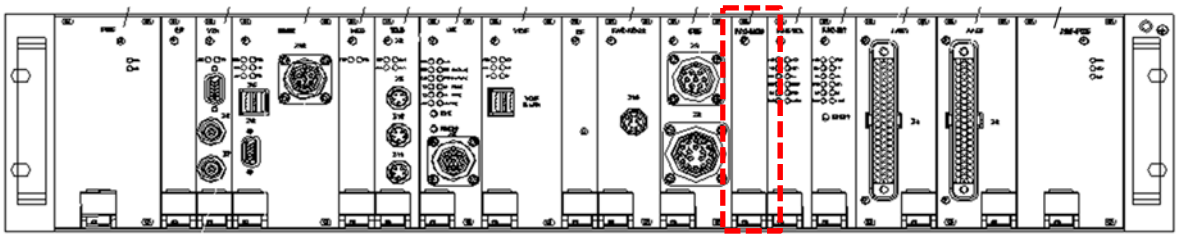

**Figure 1 –CRA- PAC Board Location**

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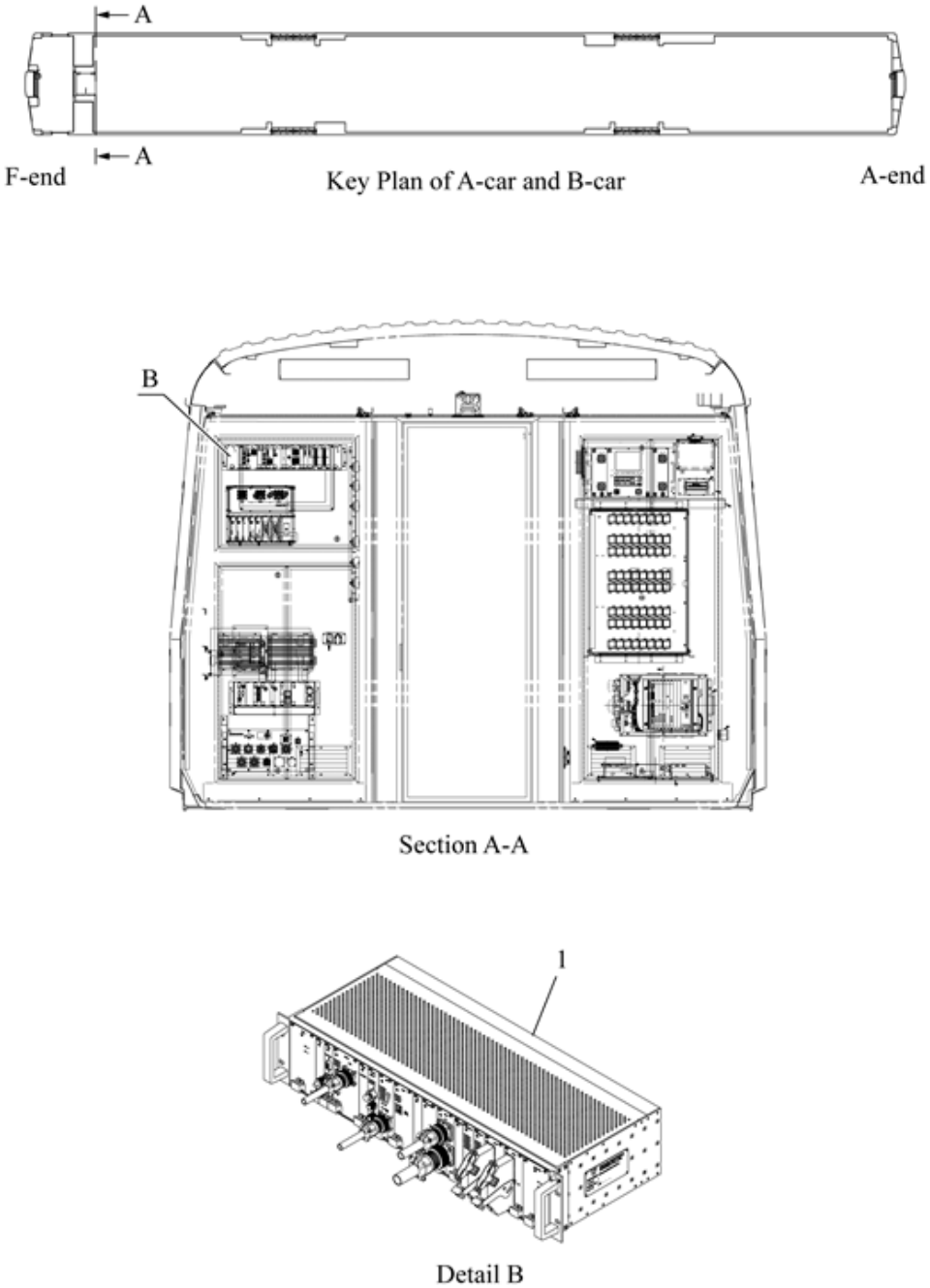
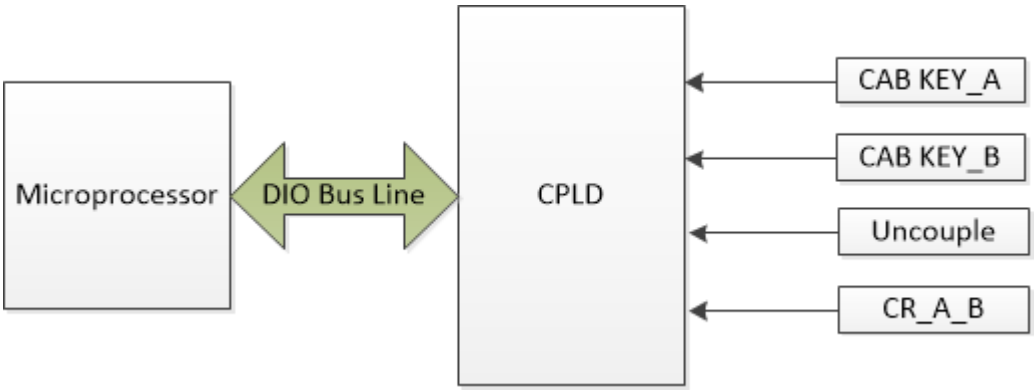



**Figure 2 – CRA Location**

## 3.2 SOFTWARE UPDATE DESCRIPTION

### 3.2.1 Algorithm Block Diagram




### 3.2.2 CAB Latch circuit change issue

- In case that the CAB Latch is changed, the communication fault is occurred because the Cab signal input is duplicated to both side of MA/MB.

### 3.2.3 Software update for CAB Latch change

1) The exceptional condition to process simultaneously input the CPLD CAB key

- If the all of CAB key A and B DIO input is ‘1’, CAB key is not recognized following the exceptional condition.

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2) The algorithm change regarding to CAB key recognition time of CRA-PAC software.

- The existing CAB key recognition is using 200ms chattering time to Cab key activation. But the timer interrupt is added in order to monitor Cab key status for 5 seconds before starting the “inauguration”.

3) Add the stabilization code for TL line communication after recognizing CRA-PAC CAB key and starting “inauguration”.

- To prevent TL line communication fault of CRA-PAC, added 5byte to the initial dummy data for the steady pay load data.

4) If RS 485 communication fault is occurred after CAB key recognition, the “retry” is activated for 3 times.

- The algorithm is added to retry “inauguration” after monitoring the response packet of the sub-system if the initial married communication fault is occurred by the external environment.

## 3.3 TEST AND VERIFICATION.

After updating the CRA-PAC software, follow the procedure below to verify the functionality:

1) Check the modification of the car CAB Latch circuit.

2) CRA-PAC SW update.

3) MA CAB TEST

- MA CAB Key ON → OFF
- Check CAB ON signal is remaining constant on CCP
- Check the CCP (Sub-system status)
- Operate CCP simulation
- MOP manual announcement
- DPH manual announcement.

4) MB CAB TEST

- MB CAB Key ON → OFF
- Check CAB ON signal is remaining constant on CCP

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- Check the CCP (Sub-system status)

- Operate CCP simulation

- MOP manual announcement

- DPH manual announcement.

### 3.4 VERSION CHECK

Use the MDS Software Version menu to verify the updated software versions.

- CRA-PAC : 1.0.04

## 4.TRACKING

Status of the field modification activities will be tracked and managed for car number and items completed. The progress will be reported to DTP/H-Rotem in daily basis as the modification proceeds.

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